Automated Pill Dispenser

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The Cost of Medication Non-Adherence

- Leads to missed doses, overdosing, and incorrect timing
- Increases risk of serious health complications and hospitalizations
- Results in reduced treatment effectiveness
- Objective: We aim to engineer a pill holder/dispenser that allows users to take medication at the correct time each day, including integration of proper authentication for users, caretakers, and physicians.

Project Overview

Feature List

- Rotating Pillbox Structure with Refillable Slots
- Interactive TFT Display with Touch
- Solenoid Deadbolt Lock
- Keypad
- Fingerprint Sensor
- Alarm played through Speakers

User Interface

- Touch display interface to choose task and authentication method
- Keypad authentication to either open the lid (refill) or to take medication
- Fingerprint sensor as simpler form of authentication
- Motor to release medication to a pill container that users can remove

Project Structure

- To open lid, a deadbolt is held open during a period for the user to physically lift the lid
- To release medication, upon authentication, a stepper motor turns a precise amount, pushing the next slot's pills
- The display reflects current state, and it allows users to add their data/medication times and see their stats
- Security: Biometric fingerprint data is not stored anywhere other than in external MCU

System Functionality: How it Works